FINM3013 Introduction to Financial Derivatives
Solution to Midterm Test

I. Multiple choice questions (45 points):

1. D. The payoff that must be made on the options is 200×(120−110) or $2000. The amount received for the options is 5×200 or $1000. The net loss is therefore 2000−1000 or $1000.
2. C. Suppose that S_T is the final asset price and K is the strike price/forward price. A short forward contract leads to a payoff of $K-S_T$. A long position in a European call option leads to a payoff of $\max(S_T-K, 0)$. When the payoffs are added together, it can be seen that the total position leads to a gain of $\max(0, K-S_T)$ which is the payoff from a long position in a put option. C can also be seen to be true by plotting the payoffs as a function of the final stock price.
3. A. The user of the commodity takes a long futures position. The gain on the futures is 63.50−59 or $4.50. The effective paid realized is therefore 64−4.50 or $59.50. This can also be calculated as the March 1 futures price (=59) plus the basis on July 1 (=0.50).
4. B. The price has increased by $2. Because you have a short position you lose 2×100 or $200. The balance in the margin account therefore goes down from $3,500 to $3,300.
5. C. To increase beta by 0.6 we need to go long 0.6×36,000,000/(900×250) or 96 contracts.
6. D. The value of the FRA is the value of receiving an extra 0.5×(0.08−0.06)×1000 = $10 in 2.5 years. This is $10/(1.03^5) = \$8.63$.
7. A. The six month rate is 1.5% per six months. The one year rate is 2% per six months. The one year par yield is the coupon that leads to a bond being worth par. A is the correct answer because $(3.99/2)/1.015+(100+3.99/2)/1.02^2 = 100$. The formula in the text can also be used to give the par yield as $[(100-100/1.02^2)\times 2]/(1/1.015+1/1.02^2)=3.99$.
8. C. If the rate is R we must have

$$\frac{3}{1.04}+103e^{-R\times 1}=97$$

or

$$e^{-R}=\frac{97-3/1.04}{103}=0.9137$$

so that $R = \ln(1/0.9137) = 0.0902$ or 9.02%.

9. C.
10. A. The 3-year forward price is the spot price grossed up for 3 years at the risk-free rate. It is $30e^{0.1\times 3}$ =$40.50.
11. B. The contract price is 110,000. The number of contracts is (24,000,000×5.5)/(110,000×6.0)=200
12. B. With the notation in the text, the futures rate exceeds the forward rate by $0.5\sigma^2 T_1 T_2$. In this case σ=0.011, T_1=4 and T_2=4.25 so the difference between the forward and futures price is $0.5\times 0.011^2\times 4\times 4.25=0.00103$.
13. B. The futures rate must be reduced by what is known as a convexity adjustment to get the forward rate.
14. D. If the forward price is too low relative to the spot price the trader should short the asset in the spot market and buy it in the forward market.

15. B. When yields increase bond prices decrease. The proportional decrease is the modified duration times the yield increase. In this case it is 5×0.0005=0.0025. The decrease is therefore 0.0025×1,000,000 or $2,500.

II. Problems (35 points)

1. (7 points)
   Since the futures price exceeds the forward rate, the arbitrageur should sell futures contracts at $0.9145 and buy euro forward in the same amount at $0.9127. The arbitrageur will earn 125,000(0.9145 - 0.9127) = $225 per euro futures contract arbitraged.

2. (10 points)
   The Eurodollar futures contract price of 89.3 means that the Eurodollar futures rate is 10.7% per annum with quarterly compounding. This is

$$4\ln(1 + 0.25 \times 0.107) = 0.1056$$

   or 10.56% with continuous compounding. The forward rate given by the 90-day rate and the 180-day rate is (10.4*180 – 10.0 *90)/90 = 10.8% with continuous compounding. This suggests the following arbitrage opportunity:
   a) Sell Eurodollar futures.
   b) Borrow 90-day money.
   c) Invest the borrowed money for 180 days.

3. (18 points)
   Please refer to lecture notes or Example 5.6 in the textbook.

Suppose that the two-year interest rates in Australia and the United States are 5% and 7%, respectively, and the spot exchange rate between the Australian dollar (AUD) and the U.S. dollar (USD) is 0.6200 USD per AUD. From equation (5.9), the two-year forward exchange rate should be

$$0.62e^{(0.07-0.05)\times 2} = 0.6453$$

Suppose first the two-year forward exchange rate is less than this, say 0.6300. An arbitrageur can

1. Borrow 1,000 AUD at 5% per annum for two years, convert to 620 USD and invest the USD at 7%. (Both rates are continuously compounded.)
2. Enter into a forward contract to buy 1,105.17 AUD for $1{,}105.17 \times 0.63 = 696.26$ USD.

The 620 USD that are invested at 7% grow to $620e^{0.07\times 2} = 713.17$ USD in two years. Of this, 696.26 USD are used to purchase 1,105.17 AUD under the terms of the forward contract. This is exactly enough to repay principal and interest on the 1,000 AUD that are borrowed ($1{,}000e^{0.05\times 2} = 1{,}105.17$). The strategy therefore gives rise to a riskless profit of $713.17 - 696.26 = 16.91$ USD. (If this does not sound very exciting, consider following a similar strategy where you borrow 100 million AUD!)

Suppose next that the two-year forward rate is 0.6600 (greater than the 0.6453 value given by equation (5.9)). An arbitrageur can:

1. Borrow 1,000 USD at 7% per annum for two years, convert to $1{,}000/0.6200 = 1{,}612.90$ AUD, and invest the AUD at 5%.
2. Enter into a forward contract to sell 1,782.53 AUD for $1{,}782.53 \times 0.66 = 1{,}176.47$ USD.

The 1,612.90 AUD that are invested at 5% grow to $1{,}612.90e^{0.05\times 2} = 1{,}782.53$ AUD in two years. The forward contract has the effect of converting this to 1,176.47 USD. The amount needed to payoff the USD borrowings is $1{,}000e^{0.07\times 2} = 1{,}150.27$ USD. The strategy therefore gives rise to a riskless profit of $1{,}176.47 - 1{,}150.27 = 26.20$ USD.